

DNA Barcoding vs ddRAD Sequencing

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Summary

DNA barcoding and ddRAD sequencing answer different kinds of genetic questions. Barcoding reads a short DNA region to help identify a fish by matching it to known references. ddRAD reads many selected regions to compare genetic variation and ancestry. For tilapia, barcoding is a useful starting point for identification; ddRAD becomes useful when the question concerns population differences or possible hybrids. The choice depends on the evidence needed, and the methods can be used together.

DNA barcoding: which species is this?

DNA sequencing reads the order of DNA's chemical "letters." Barcoding reads a standard short region and compares that sequence with records from identified specimens. In fish, the usual target is **COI** (cytochrome c oxidase subunit I), a gene in mitochondrial DNA.[1] A close match to reliable references can support a species identification. Closely related species may share similar barcodes, so a match is not always decisive.

Typical workflow: tissue → extract DNA → copy COI using PCR → sequence → compare with references.

PCR makes many copies of the target region so it can be read. Sanger sequencing is commonly used to read the COI sequence.[3] The result is a short sequence for each fish, suitable for checking an identification or comparing variants at that region.

Why COI can miss a hybrid: suppose compatible species A and B cross, with A as the mother. Their first-generation offspring normally inherits A's mitochondrial COI but nuclear DNA from both parents. A COI match to A would miss B's paternal contribution. This hypothetical example assumes distinguishable parental barcodes.[3][4]

ddRAD: how do the fish differ genetically?

ddRAD means double-digest restriction-site-associated DNA sequencing. Instead of targeting one gene, it samples many DNA regions. Two enzymes cut the DNA at specific sequences, and a selected range of fragment sizes is sequenced. This reduces the amount of DNA to process; it is not whole-genome sequencing.[2]

Typical workflow: extract DNA → cut with two enzymes → tag and select fragments → amplify and sequence → compare DNA variants.

The tags identify which fish each sequence came from; they are different from the COI barcode used for species identification. Computers process the sequences and identify **SNPs** (single-nucleotide polymorphisms): single-letter DNA differences among fish. The result is a table of genetic variants across individuals. The number of useful variants depends on the samples and data quality.[2]

Researchers compare these SNP patterns to measure genetic variation among the sampled fish and investigate whether populations differ genetically.[5]

For species identification, they compare an unknown fish's pattern with patterns from several reliably identified fish of each candidate species. Differences across many SNPs can support an assignment. Without known reference fish, the analysis may reveal genetic groups but cannot reliably give them species names.[3]

Nuclear DNA is usually inherited from both parents. Comparing a fish with references for the possible parental species can therefore help assess mixed ancestry. This requires evidence across many markers; one unusual SNP alone does not establish hybrid ancestry.[4]

How ddRAD supports a species assignment

- 1. Establish the references.** Begin with multiple reliably identified fish from each candidate species. Their labels give biological meaning to the genetic comparison.
- 2. Find reliable SNPs.** Sort reads by fish, remove poor-quality reads, align them to a genome or group matching regions, then identify and filter DNA variants.[2] Software such as Stacks supports this processing.[3]
- 3. Select informative markers.** Compare DNA-letter variants, called alleles, across references. A candidate diagnostic SNP separates species in those samples; other SNPs help when an allele is more common in one species. Check that the differences persist in additional fish.[3]
- 4. Assign from many SNPs.** Principal component analysis (PCA) displays genetic similarity. Discriminant analysis of principal components (DAPC), trained on labelled reference fish, can estimate an unknown fish’s membership probabilities for those groups. These probabilities are not ancestry fractions.[6] Mixed ancestry needs a separate ancestry analysis.[4] A database match for one fragment (using BLAST) cannot replace this assessment.

For this hypothetical example, assume every reference fish has the DNA-letter pairs shown for its species.

HYPOTHETICAL INHERITANCE AT TWO NUCLEAR SNPS		
Fish	SNP 1	SNP 2
Species A parent	A/A	C/C
Species B parent	G/G	T/T
First-generation offspring	A/G	C/T

The offspring inherits one allele per parent at each SNP. Two SNPs alone cannot prove hybrid ancestry: variation within species and errors can produce misleading patterns. Use a validated pattern across many loci.

What has been tested in tilapia?

Syaifudin and colleagues used Stacks, PCA/DAPC and trees to investigate species discrimination. They tested 24 candidate SNPs with KASP, a targeted SNP assay, in 91 fish. This included discovery fish; some candidates were not consistently species-specific.[3] This was not a universal tilapia identification panel.

For a new application, test assignments on fish excluded from marker selection, model fitting and tuning, preferably from additional populations. Report uncertain and incorrect assignments. Moses and colleagues used held-out testing for tilapia population assignment;[5] species assignment needs its own validation.

Choose by the question: COI supports reference-based identification and maternal history. ddRAD adds evidence for nuclear diversity, population differences and ancestry. Using both links the maternal barcode to the broader nuclear profile.

References

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Scope. This non-systematic explainer draws on selected primary studies checked through 12 September 2026, including some abstract-only records. No new data or analyses were generated.